

**A TECHNICAL REPORT ON
STUDENT'S INDUSTRIAL WORK EXPERIENCE SCHEME (SIWES)**

**CARRIED OUT AT
NYSC PRIMARY HEALTHCARE CENTER,
SHELLENG, ADAMAWA STATE**

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CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

The Students' Industrial Work Experience Scheme (SIWES) was established in 1973 by the Industrial Training Fund (ITF) to address the gap between theoretical knowledge and practical skills among Nigerian tertiary institution graduates (ITF, 2010). Prior to its establishment, there was a growing concern among industrialists regarding the lack of practical experience among graduates, especially those in science and technology disciplines (Akinyemi, 2011). The SIWES program serves as a skills development initiative aimed at preparing students from universities, polytechnics, and colleges of education for real-world industrial environments.

This initiative was designed to enhance students' ability to apply their classroom knowledge to real-life situations, facilitate their transition to the workplace, and equip them with the necessary technical and operational skills. It provides students the opportunity to operate and interact with equipment and machines not commonly found in their schools (Oladele & Akintayo, 2013).

Before SIWES, employers found that graduates lacked job-ready competencies due to a lack of exposure to the practical side of their disciplines. This led to high unemployment and dissatisfaction with the quality of graduates being produced (Adegboye, 2012). The situation prompted the ITF to introduce the SIWES scheme in 1973/74 as a solution to the disconnect between classroom learning and workplace demands.

Initially fully funded by the ITF, the financial burden led to the fund's withdrawal in 1978. Consequently, the federal government transferred the scheme's coordination to the National Universities Commission (NUC) and the National Board for Technical Education (NBTE) in 1979. Later in 1984, the supervision and implementation of SIWES were handed back to the ITF, which resumed full management of the program in July 1985 with federal government funding (ITF, 2010).

1.2 Objectives of SIWES

The Industrial Training Fund's Policy Document No. 1 of 1973 which established SIWES outlined the objectives of the scheme as (ITF, 2010):

- i. Provide an avenue for students in Institutions of higher learning to acquire industrial skills and experience in their respective courses of study.
- ii. Prepare students for the Industrial Work situation they are likely to experience after graduation.
- iii. Expose students to work methods and techniques of handling equipment and machinery that may not be available in their Institutions.
- iv. Make the transition from school to the world of work easier; and enhance students' networks for later job placements.
- v. Provide students with an opportunity to apply their knowledge to real work situations, thereby bridging the gap between theory and practice; and vi. Enlist and strengthen Employers' involvement in the entire educational process; thereby preparing the students for employment in Industry and Commerce.

1.3 Significance of SIWES

In the earlier stage, students are graduating without any technical knowledge or working experience and this makes them to undergo further training after securing an employment. With this reason, student industrial training was established. During this program, as designed by the ITF, students are expected to get technical assistance and acquire more experience scheme in their chosen field of study and exposed them to the usage of source machines and safety precaution where relevant before the completion of their program in their various institutions.

1.4 Scope of the Report

This report focuses exclusively on the Pharmacy Unit, covering its core functions such as medication classification (prescription and non-prescription), prescription components, common pharmaceutical abbreviations, and dispensing practices. It also addresses dispensing errors, their causes, materials used in dispensing, and step-by-step dispensing procedures. Furthermore, the report includes a classification of common drug types (e.g., antibiotics, analgesics, antimalarials) and explains the difference between generic and trade drug names. The scope is limited to the Pharmacy Unit only and is based on practical experience and observation.

CHAPTER TWO

ORGANIZATIONAL PROFILE

2.1 Overview of the Organization

NYSC Primary Health Care Centre was established in the year 1995 through the joint effort of the community and the Shelleng Local Government Council to meet its health care needs. The health facility is situated in the headquarters of the Bolki Ward and the Local Government Area. Shelleng Ward was created in 1969 while the Local Government in September 1991.

NYSC PHC shares borders with Teke and Kiri-Pote PHC from the North, Jamel PHC from the South, Guyuk Cottage Hospital from the Western part. The clinic has the following settlements which include Ungwan Ibrahim, Gwala H/Senti, Gwala (GSS), Gwala (with Bank), Gadan Tumbuki, Bokayeji, Gundurimari, ATH, Noyari, Dago, Hill-Mohammed, Sefon Pegi, Sarin, Kasuwan.

NYSC Clinic is headed by Mr. Markus Yusuf as a facility manager. The facility served as referral center to other health facility within the Local Government because of the NYSC medical doctors. The health facility renders services of all minimum package activity services with total population of 22,419 and 30 bed with mattress, where they have various unit: pharmacy, accident and emergency, OPD, laboratory, labour room, post-natal, immunization, ANC, family planning, paediatric ward, female ward, male ward, waste management side, nutritional unit, record room, TB & leprosy unit and waiting area (reception). Staffs both technical and non-technical (H/9) forty nine in number.

2.3 Organizational Chart

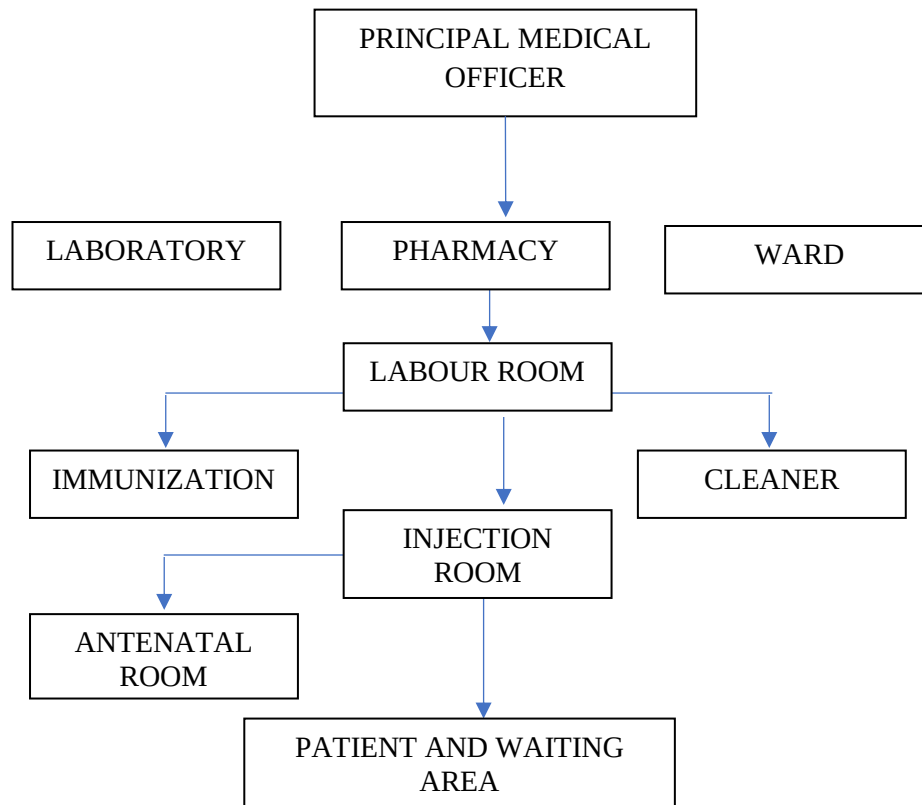


Figure 2.1: Organizational Chart

2.4 Main Activities/Services Provided

Antenatal Care (ANC), Immunization, HIV/ AIDS Services, Tuberculosis, Non-Communicable Diseases, Family Planning, Communicable Diseases, Hepatitis, Child Survival, Accidents and Emergency, Nutrition, Health Education and Community Mobilization, Maternal and newborn care.

CHAPTER THREE

NATURE OF WORK DONE

3.1 PHARMACY UNIT

During my training, I learned that prescriptions played a vital role in pharmaceutical practice. They served as formal and legal instructions issued by qualified healthcare professionals such as doctors or dentists. These prescriptions enabled us as pharmacists to accurately prepare and supply medications to patients. Each prescription typically included essential details such as the patient's name, age, address, date, prescribed drugs, dosage, and duration of treatment, along with the prescriber's signature and contact information. I became familiar with common abbreviations like o.d (once daily), b.d (twice daily), t.d.s (three times daily), and q.i.d (four times daily), which indicated how often the medication should be taken. One of the prescriptions I interpreted read "Caps Amoxicillin 500mg t.d.s x 10/7," meaning the patient had to take one capsule three times a day for ten days, totalling 30 capsules (Moffett, 2009).

In addition to dispensing, I gained practical experience with various pharmaceutical forms, including tablets, capsules, syrups, injections, and powders. I particularly found injection administration interesting, as it bypassed the digestive system and allowed for rapid drug action. I learned to administer injections through routes such as intravenous (IV), intramuscular (IM), subcutaneous (SC), and intradermal (ID), depending on the drug and desired outcome. I became proficient with tools like syringes (2ml, 5ml, 10ml), different needle gauges (21G–green, 22G–blue, 23G–purple), and scalp veins. I always followed safety protocols, especially proper disposal of syringes in safety boxes to avoid contamination. I handled and administered various medications including antibiotics like Amoxicillin and Ceftriaxone, antimalarials like Artemether and Chloroquine, and analgesics like Paracetamol and Ibuprofen.

My experience in the ward unit offered me valuable exposure to inpatient care. I attended to patients who were admitted for more than 9 to 12 hours, typically overnight, and helped document their treatment in both outpatient and inpatient registers. The wards were divided into male, female, and pediatric sections, and I attended to cases such as hypertension, diabetes, typhoid, and malaria. During ward rounds, I had the opportunity to work alongside nurses and doctors, where I administered IM and IV injections, set up and removed IV lines, stopped fluid infusions, and monitored vital signs using tools like thermometers and sphygmomanometers. This hands-on experience deepened my understanding of drug administration and patient care in a real clinical environment (Kucukarslan & Peters, 2008).

Another important aspect of my training involved learning commonly used pharmacy abbreviations such as O.D (once daily), B.D (twice daily), T.D.S (three times daily), and Q.I.D (four times daily). Familiarity with these abbreviations allowed me to better understand prescription instructions and communicate effectively with the supervising pharmacists (Babar & Scahill, 2014).

3.2 Tools and Techniques Used

During my industrial training in the Pharmacy Unit of Primary Healthcare Center Shelleng, I made use of various tools and techniques essential to pharmacy operations. These tools and practices ensured the accuracy, safety, and efficiency of dispensing medications and managing pharmacy activities (Basak & Kumar, 2012).

Counting Tray: Used to accurately count tablets and capsules for dispensing. It helped maintain hygiene and avoid direct hand contact with medications.

Measuring Spoons and Cups: Utilized to measure and administer accurate doses of liquid medications.

Permanent Marker: Used for labeling medication envelopes or containers with patient details, drug names, dosage instructions, and expiry dates.

Dispensing Envelopes: Small paper or plastic pouches used for packaging and dispensing medications to patients, keeping drugs organized and properly labeled.

3.2.2 Record-Keeping Tools

Prescription Register: A logbook used to record all dispensed prescriptions, including patient details, drug information, quantity, and date of dispensing for future reference and accountability.

Inventory Register: Used for tracking stock levels, recording incoming and outgoing drugs, and monitoring expiry dates.

3.3 Challenges Faced

- i. During my attachment in the Pharmacy Unit at NYSC Primary Healthcare Center Shelleng, I encountered several challenges that tested my adaptability and learning abilities. One of the main obstacles was interpreting illegible prescriptions due to poor handwriting by some prescribers. This made it difficult to accurately identify the prescribed drugs and dosages, increasing the risk of dispensing errors and requiring extra caution and verification.
- ii. Another challenge was the use of uncommon or incorrect abbreviations on prescriptions. Some abbreviations were unclear or used inconsistently, which initially caused confusion in understanding the exact dosing instructions. This highlighted the importance of clear communication between pharmacists and prescribers.
- iii. Limited access to digital inventory management systems also posed difficulties. Most of the record-keeping and stock monitoring were done manually, which was time-consuming and increased the chance of human error during stocktaking and expiry checks.
- iv. Furthermore, I faced challenges related to managing similar-sounding and look-alike drugs, which required heightened vigilance to prevent dispensing the wrong medication. The potential consequences of such errors underscored the critical nature of accuracy in pharmacy practice.
- v. Lastly, due to the high volume of patients, time constraints during peak hours sometimes made it difficult to provide detailed patient counseling or double-check every step thoroughly. This experience taught me the importance of time management and efficient workflow to maintain quality care even under pressure.

3.4 Solutions Applied

- i. To overcome the challenges encountered during my attachment in the Pharmacy Unit at Primary Healthcare Center Shelleng, I employed several practical solutions aimed at improving accuracy and efficiency.
- ii. For the issue of illegible prescriptions, I adopted a practice of carefully cross-checking unclear information with the prescribing doctors whenever possible. When direct communication was not feasible, I consulted more experienced pharmacists to verify the prescription details, minimizing the risk of dispensing errors.
- iii. To address the confusion caused by uncommon or incorrect abbreviations, I took the initiative to study and familiarize myself with standard pharmaceutical abbreviations and terminology. I also created a quick reference guide to assist in interpreting prescriptions more confidently and accurately.

- iv. Given the manual nature of inventory management, I helped implement a more organized system for stock monitoring by systematically labelling medicines with clear expiry dates and arranging drugs using the FEFO (First Expired, First Out) principle. This reduced the likelihood of dispensing expired drugs and made stock rotation more efficient.
- v. When dealing with look-alike and sound-alike medications, I exercised extra caution by double-checking drug names and packaging before dispensing. I also advocated for clear labelling and segregation of similar drugs on shelves to prevent mix-ups.
- vi. To manage time constraints during busy periods, I improved my workflow by preparing commonly prescribed medications in advance, and assisting pharmacists in organizing the dispensing area for faster access to frequently used drugs. Additionally, I learned to prioritize critical tasks while ensuring that essential patient counseling was delivered concisely but effectively.

Through these solutions, I was able to enhance my competence in pharmacy operations and contribute positively to the overall efficiency and safety of the dispensing process.

CHAPTER FOUR

LEARNING OUTCOME

4.1 Skills Acquired

During my attachment in the Pharmacy Unit at Primary Healthcare Center Shelleng, I developed both technical and interpersonal skills essential for effective pharmacy practice. Technically, I improved my ability to accurately interpret prescriptions, measure and dispense medications, and manage pharmaceutical inventory. I gained hands-on experience using dispensing tools such as counting trays and dosing spoons, as well as labeling and packaging medications. On the soft skills side, I enhanced my communication skills through patient counseling and collaboration with healthcare professionals. I also developed attention to detail, critical thinking, and problem-solving skills necessary to prevent and resolve dispensing errors efficiently.

4.2 Knowledge Gained

The attachment broadened my knowledge of Pharmacology, Drug classifications, and the importance of accurate prescription interpretation. I learned about the uses, side effects, and trade versus generic names of common drug classes including antibiotics, analgesics, anti-malarials, and antiretrovirals. Furthermore, I gained insight into the dispensing process, understanding the six critical steps from receiving prescriptions to patient consultation. I also acquired knowledge about common dispensing errors, their causes, and strategies to minimize them, which is vital to ensuring patient safety.

4.3 Work Ethics and Professionalism

My experience at the pharmacy unit significantly shaped my understanding of professional conduct in healthcare settings. I observed the importance of maintaining confidentiality, accuracy, and accountability when handling prescriptions and patient information. Punctuality, responsibility, and respect for colleagues and patients were reinforced as key workplace values. I learned to work under pressure with diligence and maintain ethical standards even when faced with time constraints. This attachment deepened my appreciation of the pharmacist's role in upholding patient safety and trust.

4.4 Application of Academic Knowledge

The attachment provided a practical platform to apply the theoretical knowledge I acquired in school. Concepts from Pharmacology, Pharmaceutical Calculations, and Drug classifications were directly relevant and useful when interpreting prescriptions and dispensing medications. The principles of dosage and drug interactions helped me understand the importance of accuracy in medication preparation. Moreover, the ethical guidelines and communication skills emphasized during academic training were essential during patient interactions and teamwork. This real-world experience bridged the gap between classroom learning and professional pharmacy practice, enhancing my readiness for future roles.

CHAPTER FIVE

SUMMARY CONCLUSION, AND RECOMMENDATIONS

5.1 Summary of Findings

The SIWES attachment at the Pharmacy Unit of Primary Healthcare Center Shelleng, provided valuable insights into the practical workings of a healthcare pharmacy. Key findings revealed that accurate prescription interpretation, proper medication dispensing, and effective inventory management are critical to ensuring patient safety and treatment efficacy. The experience highlighted common challenges such as illegible prescriptions and potential dispensing errors, underscoring the importance of vigilance and communication in pharmacy practice. Additionally, I gained an appreciation for the role of pharmacists in patient counseling and education.

Overall, the industrial attachment was a highly beneficial learning opportunity that bridged the gap between academic knowledge and real-world pharmaceutical practice. It enhanced my technical skills, professional attitude, and understanding of the responsibilities involved in the pharmacy profession. The supportive environment and guidance from experienced staff contributed significantly to my professional growth.

5.2 Recommendations for Future Students

- i. Future students undertaking SIWES in pharmacy should prioritize active participation and seek clarification whenever uncertainties arise, especially regarding prescription details and medication handling. Building strong communication skills is essential for effective collaboration with pharmacists and patients. Additionally, maintaining meticulous attention to detail during dispensing processes can reduce errors and improve patient outcomes.
- ii. To maximize the learning experience, students are encouraged to familiarize themselves with common pharmaceutical abbreviations and drug classifications before the attachment. They should also take initiative in learning about inventory management and patient counseling practices.
- iii. Furthermore, organizing workshops on common pharmacy errors and patient communication strategies would equip students and staff with updated knowledge and skills, promoting a safer and more effective dispensing environment.

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